UNIVERSITY OF WOLLONGONG AUSTRALIA

School of Computing and Information Technology

Individual assignment (20%)

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Supporting document to explain BPMN model assumption

This BPMN model specifies the library's private "Book delay handling" process and its interactions with external stakeholders such as "Customer", "Academic registry department", "Personnel unit", "Bad payment register", and "Debt collection agency". The process is triggered the morning after the book's due date when an overdue is detected (timer start event). From there, the library first notifies the borrower (email/ letter or optional phone call) and follows up after 7 days if no response. The follow-up process depends on the borrower's response: if the borrower replies, a librarian will record the book return, assess fees (including the book cost and a late fee of $0.50 per day accruing from the detection date), and request payment; if there is no response, the case will be escalated according to policy. Message flows are attached to concrete send/receive tasks such as reminders to the "Customer"; requests to suspend or lift to the "Academic registry department" or "Personnel unit"; register bad payer; and contact the "Debt-collection agency" for outstanding fees, maintaining proper separation between the private process and external participants. A dedicated event sub-process interrupting handles forced closure after 60 days from initial overdue detection; on trigger, the case is closed, and the borrower may be referred to the bad payer register, and further collection steps as required.

Assumptions for this model are:

1. Late fee accrual — A late fee of $0.50 per day accrues from the detection date (the morning after the due date when the overdue is identified by the system).
2. Warning/Follow-up proportions — During the warning or notification phase, empirical outcome proportions of 60%/30%/10% are recorded as text annotations for transparency. Follow-up occurs 7 days after the first reminder if there is no response from the customer.
3. Payment request outcomes — When the manager requests payment after fees are posted:

- ***Case 1 – Book returned***: Refund book cost (previously book replacement cost charged). If all fees are fully paid → lift suspension and close case; if not fully paid → only refund book cost, the case remains open, and it is either resolved later or force-closed at 60 days from initial detection.
- ***Case 2 – Book not returned***: Case remains open and is either resolved by return/payment later or force-closed at 60 days from initial detection.